

Learning from Click-Logs on EB-NeRD and MIND

CS4.406 Information Retrieval & Extraction — Assignment 2. Every number is read from a JSON written by the code that produced it.

1 What was built

A two-stage retrieve-then-rank pipeline on EB-NeRD and MIND, continuing A1's lexical and semantic retrieval. **Stage one** is A1's candidate generator over a 7-day live universe — BM25 at the per-corpus tuned (k_1 , b), cosine to a mean-pooled history vector, fused by reciprocal rank, with already-read articles dropped. **Stage two** is a LightGBM LambdaMART re-ranker (Q2 Option A) over 32 behavioural features.

Why a GBDT. The design matrix is heterogeneous and full of *structural* nulls: MIND has no `session_rank` because MIND ships no sessions, a cold user has no `hours_since_last_click` because they have no prior clicks, and a BM25 score is not on the scale of a cosine. Axis-aligned splits take all three without imputation or rescaling, and each imputation choice would have been a confound in the ablations.

dataset	features	train rows	trees	fit
EB-NeRD	32	2,226,294	257	18.4s
MIND	29	4,621,015	13	4.8s

Features follow the standard learning-to-rank families: user-only (history depth, activity, category breadth, prior dwell), article-only (CTR, decayed and rolling popularity, freshness), match (BM25, three embedding similarities, category affinity), and serving context (session position, device). Two boundaries are drawn deliberately.

Availability, not usefulness, decides what ships. The Codabench split has no labels, so rolling *click* counters cannot be computed there while rolling *exposure* counters can — candidate lists are published. The families split into shipped (submittable) and production (what a live feature store would hold); the gap between them is reported as the cost of the evaluation setup, never as a leaderboard result.

The behaviour window is enforced per row, not per split. A1's tests check the store at split granularity, which cannot see whether a session feature read a row thirty seconds in the future. `tests/test_behaviour_window.py` tests the property by construction — assemble the matrix, delete the future, assemble again, require the survivors to match — and it found two real defects: an expanding mean that used row order as a tiebreak (EB-NeRD stamps to the second, so a visit beginning at the same instant could contribute its dwell to

the row being scored), and a user-lookup that clipped an out-of-range id onto a real, different user.

1.1 Stage-one recall is the ceiling on everything downstream

dataset	K	recall@K	impressions with no click retrieved
EB-NeRD	100	0.1189	88.0%
EB-NeRD	200	0.2074	79.2%
MIND	100	0.0441	93.6%
MIND	200	0.0676	90.3%

Stage two cannot rank what stage one never retrieved, which is why the in-impression numbers below — the question both leaderboards actually score — are the ones that carry.

2 Baseline, improvement, ablation, CI

2.1 Re-ranking, in-impression — what both leaderboards score

Before is A1's best single signal on identical rows, which is the honest baseline for what *learning a combination* buys rather than what it buys over nothing.

dataset	A1 emb (before)	AUC	MRR	nDCG@5	nDCG@10	production AUC
EB-NeRD	0.5453	0.7302	0.4977	0.5615	0.6041	0.7812
MIND	0.6368	0.6452	0.3110	0.3414	0.4021	0.6990

2.2 The starter baseline, reproduced

NRMS-docvec from `ebnerd-benchmark`, ported to PyTorch layer for layer: news encoder, multi-head self-attention with no output projection or residual (the 2019 formulation, not a transformer block), additive attention pooling, dot-product scorer, softmax cross-entropy over one positive and four sampled negatives. The starter is TensorFlow and pins `polars==0.20.8`, `numpy<1.26.1`, `torch<2.3`, which will not co-install with this stack or run on this card; a frozen parallel environment would reproduce the code while making every number incomparable, so the architecture is ported and the inputs and metrics are shared.

2.3 The improvement: freshness / rolling priors

A1's popularity is computed once at the split boundary and held fixed. On a news corpus that leaves **84.2%** (EB-NeRD) and **54.9%** (MIND) of candidate slots at a prior click count of exactly zero — the same defect that put A1's MIND popularity submission at 0.4900, below chance. Kept rolling, strictly before each request, the dead fraction falls to **3.7%** and **3.7%**.

dataset	variant	AUC	Δ AUC (paired, 95% CI)
EB-NeRD	- user	0.7288	-0.0015 [-0.0019, -0.0011]

EB-NeRD	- article	0.5850	-0.1453 [-0.1465, -0.1439]
EB-NeRD	- match	0.7023	-0.0279 [-0.0289, -0.0271]
EB-NeRD	- context	0.7157	-0.0145 [-0.0151, -0.0140]
EB-NeRD	+ rolling (the improvement)	0.7812	+0.0509 [+0.0501, +0.0517]
EB-NeRD	+ grey (Q9, must not ship)	0.7304	+0.0002 [-0.0002, +0.0006] (<i>spans 0</i>)
MIND	- user	0.6473	+0.0021 [+0.0010, +0.0034]
MIND	- article	0.6498	+0.0046 [+0.0034, +0.0058]
MIND	- match	0.5173	-0.1279 [-0.1311, -0.1252]
MIND	- context	0.6456	+0.0004 [-0.0009, +0.0016] (<i>spans 0</i>)
MIND	+ rolling (the improvement)	0.6990	+0.0538 [+0.0522, +0.0557]
MIND	+ grey (Q9, must not ship)	0.6456	+0.0004 [-0.0009, +0.0016] (<i>spans 0</i>)

Two axes are kept apart, because reporting one number for an architecture change and a feature change at once is what Q3.3 forbids. The same feature change measured on the neural baseline:

dataset	model	AUC	Δ AUC (paired, 95% CI)
EB-NeRD	NRMS (baseline)	0.6041	—
EB-NeRD	+ freshness, bounded	0.6097	+0.0056 [+0.0046, +0.0068]
EB-NeRD	+ freshness, unbounded	0.6996	+0.0926 [+0.0913, +0.0939]
MIND	NRMS (baseline)	0.6467	—
MIND	+ freshness, bounded	0.6513	+0.0045 [+0.0029, +0.0063]
MIND	+ freshness, unbounded	0.4919	-0.1508 [-0.1536, -0.1479]

The bounded and unbounded forms of the neural head are both reported because the comparison is the result. An unbounded branch added to a dot product can explain the slate-softmax loss on its own and starve the tower that must generalise to a full candidate list: it wins large where freshness genuinely dominates and loses large where content does. Bounding the contribution to $\alpha \cdot \tanh(h)$, with α learned, is positive on both.

2.4 The objective arc

One config flag spans pointwise to listwise. Early stopping watches nDCG for every objective here; letting the stopping rule follow the objective confounds the comparison, and on MIND that showed as 148 trees against 13.

dataset	objective	Δ nDCG@10 vs pointwise (95% CI)
EB-NeRD	rank_xendcg	-0.0022 [-0.0027, -0.0017]
EB-NeRD	lambdarank	+0.0070 [+0.0064, +0.0077]
MIND	rank_xendcg	-0.0020 [-0.0030, -0.0010]
MIND	lambdarank	-0.0047 [-0.0058, -0.0035]

The predicted listwise > pointwise ordering holds on EB-NeRD and inverts on MIND. LambdaRank's truncation explains about two thirds of it: past `lambdarank_truncation_level` a swap contributes no gradient, and the optimum tracks list length — 30 for EB-NeRD's median-9 lists, monotonically deeper for MIND's median-23, max-295 ones.

Effect sizes are small where the intervals are tight. With 244,647 and 73,152 test impressions, thousandths of a point clear significance. That is a statement about sample size, not about what a user would notice.

3 Extended evaluation

dataset	model	AUC	nDCG@10	ILD@10	novelty@10	coverage@10
EB-NeRD	shipped	0.7302	0.6041	0.769	36.62	0.200
EB-NeRD	content-only	0.7078	0.5819	0.766	36.77	0.196
EB-NeRD	article-log-only	0.6251	0.5106	0.772	36.20	0.204
MIND	shipped	0.6452	0.4021	0.903	28.55	0.057
MIND	content-only	0.6457	0.4051	0.902	29.09	0.057
MIND	article-log-only	0.4930	0.2870	0.944	30.83	0.068

Diversity is intra-list distance in the embedding space rather than over the category taxonomy: MIND's taxonomy is 18 labels wide and EB-NeRD's is a different taxonomy, so a category index is not comparable across the two. Novelty is scored against the click distribution of the window *before* the split — in-window popularity would make a system look novel for surfacing what later turned out to be a hit.

dataset	slice	impressions	AUC	nDCG@10
EB-NeRD	cold	25,105	0.7287	0.5898
EB-NeRD	warm	219,542	0.7304	0.6057
EB-NeRD	head	49,121	0.7166	0.5374
EB-NeRD	tail	195,526	0.7337	0.6208
MIND	cold	12,982	0.5798	0.3864
MIND	warm	60,170	0.6593	0.4055
MIND	head	19,915	0.6566	0.4482
MIND	tail	53,237	0.6409	0.3849

3.1 Two lecture claims, measured

Position bias is not in this data. The naive CTR-by-position curve decays sharply on both datasets, but both logs are near-single-click, so per-slot CTR is $\approx 1/L$ by arithmetic and position k is reachable only by lists longer than k . Holding list length fixed, the curve is flat.

dataset	raw CTR decay	ratio at fixed L	max deviation from 1.0
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EB-NeRD	0.1555	[0.8359, 1.1599]	2.6 SE
MIND	0.1546	[0.7708, 1.1952]	3.9 SE

Neither dataset stores candidates in rendered order. Inverse propensity weighting was therefore dropped as the improvement — it would correct a confound that is absent — and position stays quarantined on evidence rather than caution.

Head versus tail. The claim is that memorised article statistics are unbeatable on the head and empty on the tail. Measured with the arms defined at column level:

dataset	slice	content AUC	article-log AUC	winner
EB-NeRD (two-stage)	head	0.9168	0.7411	content
EB-NeRD (two-stage)	tail	0.9808	0.8976	content
EB-NeRD (in-impression)	head	0.7025	0.5250	content
EB-NeRD (in-impression)	tail	0.7091	0.6503	content
MIND (two-stage)	head	0.9083	0.9927	article-log
MIND (two-stage)	tail	0.9224	0.9686	article-log
MIND (in-impression)	head	0.6285	0.3459	content
MIND (in-impression)	tail	0.6521	0.5481	content

The answer depends on the candidate set. In-impression, content wins all four slices and the claim is falsified; through the cascade MIND inverts, because stage one has already discarded the articles a popularity prior would rank wrongly. The claim is a property of the candidate set, not of the data — on 420 impressions, the thinnest evidence here. A confound it cannot escape: the slice is defined by exposure count and the article-log arm is built from exposure counts, so conditioning removes that variance inside the slice. The arms compare *within* a slice, never *across* one.

4 Serving and scale

dataset	ANN index	BM25 index	article vectors	feature store (disk)	resident
EB-NeRD	4.91 MB	2.36 MB	60.76 MB	125.68 MB	237.26 MB
MIND	33.36 MB	12.89 MB	95.56 MB	53.2 MB	1006.13 MB

dataset	p50	p95	p99	cand. gen	re-rank	fetch share of re-rank
EB-NeRD	0.77	0.81	0.84	0.462	0.313	16%
MIND	3.52	4.23	4.80	3.428	0.13	42%

dataset	p99	SLA met	q/s (1 thread)	\$ / 1000 queries
EB-NeRD	0.84 ms	yes	1288.8	\$0.000002
MIND	4.80 ms	yes	281.0	\$0.000010

The per-box figure assumes linear scaling across cores and is stated as an upper bound: A1 measured $110 \rightarrow 467$ q/s going from 1 to 48 leaves, $4.23\times$ for $48\times$ the threads.

5 Where it breaks at $10\times$

what breaks	why	the fix
the frozen feature store	popularity fixed at the split boundary already leaves 84.2% of EB-NeRD candidate slots at a zero prior click count, and that fraction grows with catalogue churn	the rolling counters measured here (+0.0509 AUC), but served from a streaming aggregate rather than a batch job
exact ANN search	A1 measured the flat-vs-HNSW crossover at $N \approx 8,000$; the live universe passes it at $10\times$ the window	HNSW — $9.4\times$ throughput at 125K articles for 96.5% fidelity, whose 42 s single-threaded build becomes the new bottleneck
the BM25 matmul	H·TF stops fitting at $10\times$ users $\times$ $10\times$ vocabulary	blocked product, or WAND once the <i>universe</i> grows $\sim 5\times$. A1 measured that the query path breaks at $10\times$ the universe, not $10\times$ the users
the re-rank feature join	the cascade build OOMed at $K=200$ over 200k impressions; peak RSS is linear in rows, ~ 7.4 GB per 5k impressions	chunked scoring, which is what makes the 205,925,868-slot EB-NeRD submission possible at all
thread oversubscription	LightGBM and polars each default to every core; two concurrent steps request twice the machine. Measured at load 77 on 48 cores, an ablation that takes 139 s had not finished in 59 minutes	explicit thread caps — the same work finished $25\times$ faster at 12 threads
single node	one box, no replication	the one item a single machine cannot measure, stated as a gap rather than estimated

6 What the measurements changed

- Position bias was assumed and is absent.** The raw curve decays $6.4\times$ and is an artifact of list length. IPW was dropped as the improvement on that evidence.
- The dwell feature's denominator beat the dwell.** A rolling click counter arrived by accident as an intermediate and turned out to be the strongest feature in the model; it was then built properly, for both datasets, as the shipped improvement.
- Each training candidate set wins on the framing it matches.** Training stage two on retrieved sets needs clicked articles unioned back in, and the union *is* the label — 0.99 AUC on its own retrieved sets, below random in-impression.

4. **Features unavailable at serving time are worth nothing here** (+0.0002 AUC, CI spans zero), which is consistent with the position-bias finding rather than independent of it.